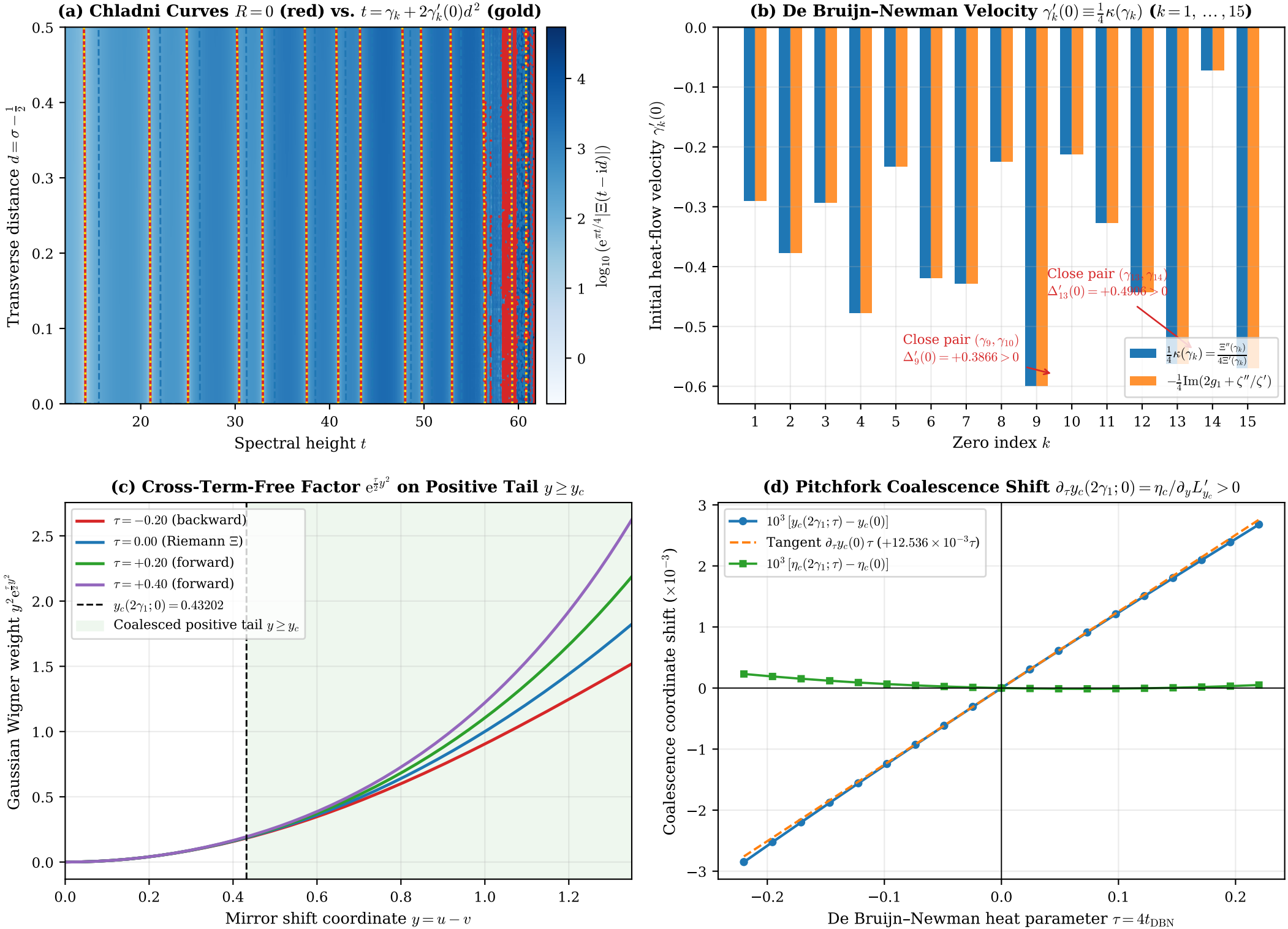

(a) Chladni Curves $R=0$ (red) vs. $t=\gamma_k+2\gamma_k'(0)d^2$ (gold)
Transverse distance $d=\sigma-\frac{1}{2}$
Spectral height t
$\log_{10}(e^{\pi t/4}|\Xi(t-id)|)$
(b) De Bruijn–Newman Velocity $\gamma_k'(0)\equiv\frac{1}{4}\kappa(\gamma_k)$ $(k=1,\ldots,15)$
Initial heat-flow velocity $\gamma_k'(0)$
Zero index k
Close pair $(\gamma_{13},\gamma_{14})$
$\Delta_{13}'(0)=+0.4906>0$
Close pair (γ_9,γ_{10})
$\Delta_9'(0)=+0.3866>0$
$\frac{1}{4}\kappa(\gamma_k)=\frac{\Xi''(\gamma_k)}{4\Xi'(\gamma_k)}$
$-\frac{1}{4}\mathrm{Im}(2g_1+\zeta''/\zeta')$
(c) Cross-Term-Free Factor $e^{\frac{\tau}{2}y^2}$ on Positive Tail $y\geq y_c$
Gaussian Wigner weight $y^2e^{\frac{\tau}{2}y^2}$
Mirror shift coordinate $y=u-v$
$\tau=-0.20$ (backward)
$\tau=0.00$ (Riemann Ξ)
$\tau=+0.20$ (forward)
$\tau=+0.40$ (forward)
$y_c(2\gamma_1;0)=0.43202$
Coalesced positive tail $y\geq y_c$
(d) Pitchfork Coalescence Shift $\partial_\tau y_c(2\gamma_1;0)=\eta_c/\partial_y L_{y_c}'>0$
Coalescence coordinate shift $(\times10^{-3})$
De Bruijn–Newman heat parameter $\tau=4t_{\mathrm{DBN}}$
$10^3\,[y_c(2\gamma_1;\tau)-y_c(0)]$
Tangent $\partial_\tau y_c(0)\,\tau$ $(+12.536\times10^{-3}\tau)$
$10^3\,[\eta_c(2\gamma_1;\tau)-\eta_c(0)]$